Alex Eftimiades

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Work Experience

Penguin Random House Remote
Applied ML Scientist Nov 2022 - Present

- Built constrained MMM attribution and optimization models projecting ROI across hypothetical spend scenarios.
- Designed constrained neural nets (JAX) encoding spend response shape and monotonicity priors to correct platform attribution bias; used xgboost where supported.
- Models projected to save $8M/year via spend constraints and allocation.
- Presented modeling approach at Data Science Salon (2023) and ANA Marketing Conference (2025).
- Applied difference-in-differences with controls matched on genre and days since publication to estimate incremental lift on channels with limited spend data.
- Built team adopted Claude Code hook using 12 regex classifiers and an LLM judge to block error swallowing patterns; catches 2 violations per session on average.
- Built autonomous agent that monitors production failures via Slack, files Claude Code merge requests, and verifies fixes against reruns and production data.
- Supervised a rotational data science intern and mentored other data scientists.
- Built and deployed AI driven ad generation and monitoring pipeline using Python and Kubernetes. Ads deployed to Facebook, Instagram, and Google.
- Built Databricks and Claude Code infrastructure linking Git revisions to Snowflake data, model artifacts, and pipeline runs, enabling reproducible parallel experimentation and establishing the team's standard for code and data lineage.

SPAR AI — Research Fellow Sep 2026 - Present
Investigating topological features of LLM activations for deception detection against linear probe baselines.

Trekkit LLC Remote
Founder Jan 2024 - Present

- Built and launched several financial planning apps on decisiondough.com (SST/AWS).
- Built and launched Offensive Weather to Google Play Store and Apple App Store (SST/AWS).

FINRA Remote
Lead Data Scientist June 2019 - Nov 2022

- Led deployment of NLP models in production (Docker, Lambda/AWS), reducing costs by 80%.
- Led R&D and open sourced the Model Validation Toolkit — a research library implementing constrained interpretable neural nets, Bayesian credibility analysis, and divergence-based drift detection.
- Developed supervised and unsupervised models to identify insider trading (xgboost; 96% AUC), market manipulation (dbscan), fraud (bayesian analysis), and triage external communication (xgboost, sklearn, BERT).
- Mentored data scientists and led workshops on machine learning and statistics principles and practices.
- Gave internal talks on: software engineering for data scientists, countering sample bias, measuring model drift, thresholding, normalizing flows.
- Developed and conducted technical interview process and brought on 7 data scientists.
- Promoted twice in three years.

Deepsig Arlington, VA
Machine Learning Engineer January 2019 - March 2019

- Designed and implemented deep learning based signal detector and classifier.
- Compared deep learning approaches benchmarked against classical clustering algorithms for signal identification and classification.
- Gave talk on semi-supervised learning.

Catalist LLC Washington DC
Analytics Engineer February 2018 - January 2019

- Optimized, parallelized, and deployed NLP Keras model.
- Wrote SQL parser that refactored over one million lines of legacy SQL scripts.
- Designed and wrote data processing pipeline for election results as they became available the night of the election.
- Contributed code to Keras (fixed tokenizer).

Comsol Burlington, MA
Developer February 2016 - May 2017

- Researched models and techniques to simulate physical phenomena of interest to engineers and scientists.
- Implemented and documented algorithms used for numerical simulations and user interfaces in Java.
- Helped customers create and optimize simulations.

Skills

Programming Languages:	Python, Bash, SQL
Frameworks:	Jax/Pytorch, Numpy/Scipy, Cython, Pandas, Scikit-learn
Tools:	Git, Vim, AWS, Kubernetes, Jupyter, Plotly, Docker

Projects

Fitness Mortality Calculator (Apr 2026) *Python* https://aeftimia.github.io/fitness-mortality/
Evidence based web app modeling mortality risk from VO2 max and grip strength using published hazard ratios and SSA life tables.

Model Validation Toolkit (Dec 2021) *Python* https://github.com/FINRAOS/model-validation-toolkit
Open source ML validation library covering interpretable neural networks, Bayesian credibility analysis, drift detection, and sensitivity analysis.

Discrete Exterior Calculus Framework (May 2014) *Python, Cython, CUDA*
https://github.com/aeftimia/kahler
GPU parallelized discrete exterior calculus framework for Hermitian manifolds; extends PyDEC with Cython/CUDA and supports arbitrary metrics, periodic boundary conditions, and cohomological operators.

Hexchat (July 2013) *Python* https://github.com/aeftimia/hexchat
Custom TCP-over-XMPP tunneling protocol for Tor, supporting multi account JID rotation and ordered delivery across arbitrary chat relays to bypass censorship bandwidth limits.

Awards and Publications

Leveraging Data Science to Optimize Retail Media Campaigns ANA Marketing Conference
Presented research on mixed market models at ANA Marketing Conference. Nov 2025

Advertising at Scale Data Science Salon
Presented work on automated advertising and A/B testing at Data Science Salon. June 2023

Introducing the Model Validation Toolkit Open Source Data Science (ODSC) East 2022
Gave talk at ODSC East introducing open source toolkit developed internally at FINRA. April 2022

Enhancing the Three-Dimensional Structure of Adherent Gingival Fibroblasts and Spheroids via a Fibrous Protein-Based Hydrogel Cover Cells Tissues Organs
Published with biologists at the American Dental Association Foundation. August 2016

Kahler: An Implementation of Discrete Exterior Calculus on Hermitian Manifolds
arXiv:1405.7879
Independent research and implementation of a GPU parallelized finite elements framework for Hermitian manifolds. May 2014

Education

UMBC Catonsville, MD
BS Physics (Minor in Mathematics) 2013 - 2015

BlueDot Impact September 2026
AGI Strategy